MIL-L-43579C
7 October 1982
SUPERSEDING
MIL-L-43579B
2 April 1976

MILITARY SPECIFICATION

LINER, TROUSERS, CHEMICAL PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for a cotton sateen trouser liner.

1.2 Classification. The liner shall be of one type in the following sizes (see 6.2).

Schedule of sizes

X-Small
Small
Medium
Large
X-Large

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

FEDERAL

L-P-398	-	Plastic Molding Material. Styrene-Butadiene
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
V-T-276	-	Thread, Cotton
V-T-280	-	Thread, Gimp, Cotton, Buttonhole
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
DDD-T-86	-	Tape, Textile; Cotton, General Purpose (Unbleached, Bleached or Dyed)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-10296	-	Cloth, Sateen, Cotton

STANDARDS

FEDERAL

FED-STD-191	-	Textile, Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

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3.2 Materials (see 6.4).

* 3.2.1 Basic material. The basic material shall be cotton sateen cloth conforming to class 1 of MIL-C-10296 with the additional requirement that it shall show a minimum dynamic absorption of 60 percent when tested as specified in 4.2.1. The material shall be dyed Olive Green 107.

3.2.2 Cuff tubing. The fabric for the cuff shall be a true 1 by 1 rib knit, seamless, cotton tubing conforming to the following requirements when tested as specified in 4.2.1.

Yards per pound:	Minimum 9, maximum 12
Wales per inch:	Minimum 16
Courses per inch:	Minimum 20
Flat width, inches:	Minimum 3

3.2.2.1 pH. The pH value of the water extracted from the finished cloth shall be not less than 5.0 or more than 7.0 when tested as specified in 4.2.1.

* 3.2.2.2 Color. The cloth shall be dyed Olive Green 109 and shall approximate the color of the base material. The use of sulfur dyes and dyes containing elementary sulfur or components capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more labile sulfur than shown by the standard sample, when tested as specified in 4.2.1.

* 3.2.2.3 Colorfastness. The dyed finished cloth shall show fastness to laundering, perspiration and crocking equal to or better than the base material.

* 3.2.2.4 Non-fibrous material. Starch and protein content including chloroform-soluble and water-soluble material in the finished cloth shall not exceed 3.0 percent when tested as specified in 4.2.1.

3.2.3 Tape. The tape for liner shall be type I, class 3, 1/2 inch wide conforming to DDD-T-86 except the colorfastness to light shall not apply. The tape shall be dyed Shade S, Cable No. 66519.

3.2.4 Buttons. The buttons shall be type II, class D, style 26, 30-line, conforming to V-B-871. The buttons shall have a dull finish and shall match Shade Olive Green BP, Cable No. 62016.

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3.2.5 Toggle, plastic. The plastic toggle shall be fabricated from polystyrene rodstock. The plastic material used in the rodstock shall meet the requirements of either type II or III, grade B of L-P-398. The material shall have a dull finish and shall match Shade Olive Green BP. The toggle shall conform to the measurements specified in figure 2.

3.2.6 Thread. The thread for seaming and stitching shall be cotton, type IA3, Olive Drab S-1, C.A. 66022, ticket No. 30 and 50, 3 ply conforming to V-T-276.

* 3.2.6.1 Colorfastness. The dyed thread shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering (after 3 cycles).

3.2.7 Gimp. The cotton gimp for reinforcing buttonholes shall be Olive Drab S-1, C.A. 66022, conforming to type I or II, size No. 8 of V-T-280.

* 3.2.7.1 Colorfastness. The dyed gimp shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample. When no standard sample is available, the dyed gimp shall show "good" fastness to laundering (after 3 cycles).

* 3.2.8 Labels. Each liner shall have a size label, an identification label, and an instruction label or a combination size, identification and instruction label conforming to type VI of DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.2.8.1 Identification label. The identification label shall conform to class 1.

* 3.2.8.2 Size label. The size label shall conform to class 2 and shall contain the following information for each respective size. Description information shall be printed in letters not less than a minimum of 10-point.

X-Small	Medium
Waist: Below 27 inches	Waist: From 31 to 35 inches
Stock No.	Stock No.
NATO size: 5869	NATO size: 7989
Small	**Large**
Waist: From 27 to 31 inches	Waist: From 35 to 39 inches
Stock No.	Stock No.
NATO size: 6979	NATO size: 8999

X-Large

Waist: From 39 to 43 inches
Stock No.
NATO size: 9909

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* 3.2.8.3 Instruction label. The instruction label shall conform to class 3, and shall contain the following information:

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1. Wear a chemical protective layer when a chemical attack is imminent.

2. Wear over cotton "Shorts" or winter underwear.

3. Adjust waist size by drawing on tape and pushing toggle toward liner. Tie excess tape in such a manner so that it can be released for re-adjustment, laundering or reissue.

4. Button inside gas flap to right side of waist, then button waist button.

5. Wear waist of Liner, Trousers, Chemical Protective, over bottom of Liner, Shirt, Chemical Protective, only, at all times.

6. Wear protective socks, over cuff of Liner, Trousers, Chemical Protective, and be sure both are caught in the boot.

7. Use Buddy System, if necessary, when donning socks.

8. Do not tuck bottoms of field clothing into footwear tops.

9. Launder by prescribed laundry methods. Reimpregnate according to Field Laundry Procedures. Alternatively, reimpregnate with chloramite powder (XXCC3) from M-13 Decontamination Kit, using established procedures.

10. Follow field laundry impregnation procedures strictly. Overheated clothing may ignite spontaneously.

DO NOT REMOVE THIS LABEL

* 3.2.8.4 Combination size, identification and instruction label. The combination label shall conform to class 14.

3.3 Design. The liner has a grown on waistband, waist adjustment tape, protective flap, single button front closure, and knitted cuff bottoms.

3.4 Patterns. Standard patterns which provided an allowance of 1/2 inch for joining seams and 3/8 inch for all other seams will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns.

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3.4.1 List of pattern parts. The components of the liner shall be cut from material as specified according to the pattern parts indicated.

List of pattern parts

Material	Nomenclature	Cut parts
Cotton sateen cloth	Back	2
	Front	2
	Protective flap	2

* 3.5 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Where two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the liner. The gage of all overedge stitching, shall be 3/16 inch and the seaming row, when using safety stitching, shall be 3/16 inch from back edge of overedge stitching. The gage of stitching for double-lapped, double-stitched seams shall be 1/4 to 5/16 inch, with the outer row of stitching 1/16 inch from turned edge on outside of liner. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained so that there will be no tight or loose stitching.

* 3.5.1 Type 301 stitching. Ends of all stitchings shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching (except labels) shall overlap not less than 1/2 inch. The ends of label stitching shall be overlapped not less than 3 stitches. Thread tensions shall be maintained so that there will be no loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

* 3.5.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

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* 3.5.1.2 Type 401, 502, 503 and 515 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.5.1.1 a. and 3.5.1.1 b. 301 stitching may be used in repairing 401 stitching.

* 3.5.2 Bartacks. Bartacks shall measure 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch in width and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose or tight stitching.

3.5.3 Buttonholes. The buttonholes shall be the eyelet end taper bar type, worked over gimp with ends of the buttonhole stitching and gimp tacked. The stitching shall be caught in the cloth with the purling on the outside and the cut lengths shall be 3/4 to 7/8 inch.

3.5.4 Eyelet. Eyelets shall be sewn with the purling on the outside with a 1/8 to 3/16 inch inside diameter. The width or bight of stitching around the eyelet shall be not less than 1/16 inch. The eyelet stitching shall have at least four overlapping stitches.

* 3.6 Manufacturing operations requirements. The liner shall be made in accordance with operation requirements specified in table I. The contractor is not required to follow the exact sequence of operations listed, unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible, provided the thread does not show on the finished liner.

3.6.1 Figures. Figures 1 and 2 are furnished for information only. To the extent of any inconsistencies between the figures and the written document the written document shall govern.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the liner in strict accordance with patterns furnished which show size, directional lines and notches for proper assembly of all parts. The filling effect side of the material shall be the face. b. Lay the fronts and backs of liner so that the directional line is in the warp direction. The directional line may vary from the warp direction by not more than 2-1/2 inches on both fronts and backs. Measurements shall be taken from the top and bottom of the directional line on pattern to the selvage edge of the cloth and the difference between two measurements shall not exceed the tolerance specified. c. The waist adjustment tape shall be cut in a sufficient length to extend 3 $\pm$ 1 inches from each buttonhole or eyelet, when threaded through the toggles.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.2.2 shall be removed from production and replaced with non-defective and properly matched components.						

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Component marking.						
	Mark all components of the liner to insure a uniform size throughout the liner. Any method of marking may be used except:						
	a. Metal fastening devices.						
	b. Sew on type tickets.						
	c. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
4.	Front opening.						
	a. Turn in edge of left front opening 1/4 to 5/16 inch with the raw edge turned in and single stitch 1/16 to 3/32 inch from edge.	301	EFb-1	10-14	30-3	30-3	
	b. Turn edge of right front opening to outside 1/4 to 5/16 inch with the raw edge turned in and single stitch 1/16 to 3/32 inch from edge.	301	EFb-1	10-14	30-3	30-3	
5.	Make protective flap.						
	a. Single stitch plies of flap together along side and bottom edges. Turn and work out edges.	301 or 401	SSa-1 SSa-1	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Turn top raw edges between plies and single stitch 1/16 to 1/8 inch from edge.	301	SSc-1	10-14	30-3	30-3	
6.	Join crotch and seat seams.						
	a. Turn crotch seam of right front to the outside and stitch 1/16 to 1/8 inch from edge.	301	LSdc-3(a)	10-14	30-3	30-3	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Join crotch and seat seams. (cont'd)						
	b. Turn crotch seam of left front to inside and double-stitch 1/4 to 5/16 inch gage to the right front crotch. The top ends of stitchings shall terminate at bottom end of fly opening.	401	LSdc-3(b)	10-14	30-3	50-3	
	c. Join the seat seam with a double-lapped and double-stitched seam 1/4 to 5/16 inch gage from top of waist. The right back shall lap on left back.	301 or 401	LSc-2	10-14	30-3 30-3	30-3 50-3	
7.	Join outseams and inseams.						
	Finished appearance. The seams shall be joined with notches matching to avoid twisting and puckering. The fronts shall lap on backs.						
	a. Join outseams with a double-lapped and double-stitched seam 1/4 to 5/16 inch gage.	301 or 401	LSc-2	10-14	30-3 30-3	30-3 30-3	
	b. Join the inseams in one continuous operation with a double-lapped and double-stitched seam 1/4 to 5/16 inch gage. A 3/8 inch stagger (center to center) at crotch is permitted.	301 or 401	LSc-2	10-14	30-3 30-3	30-3 50-3	
8.	Finish waist.						
	Finished appearance. The waistband shall be smooth and flat without fullness, gathers, or pleats.						

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Finish waist. (cont'd)						
	a. Make two horizontal buttonholes for waist adjustment tape, through reinforcement pieces, as indicated by marks on pattern. Position the eyelet end of each buttonhole toward front of liner. Reinforcement piece shall be of sufficient width and length to reinforce adjustment tape outlets. The reinforcement pieces may be prestitched into position prior to making outlets.	Buttonhole		52-56 per buttonhole including tack	30-3	50-3	
	or						
	b. Make two sewn eyelets through reinforcement pieces, positioned as indicated for the eyelet end of the buttonhole.	401 or		19-24 per eyelet	30-3	50-3	
		502 or		26-32 per eyelet	50-3	50-3	
		503			50-3	50-3	
	c. Thread tape through buttonholes or eyelets with tape extending around waistband.						
	d. Bartack the waist adjustment tape through the underside of the waistband at the center back; or bartack the tape through the finished waistband.	Bartack		28 per bartack	50-3	50-3	
	e. Turn top edge of waist to outside with raw edge turned in and single stitch 1/16 to 3/32 inches. The waistband shall finish 1-1/4 ± 1/8 inches in width. The waist adjustment tape shall not be caught in the waistband stitching.	301	EFb-1	10-14	30-3	30-3	
	f. Thread adjustment tape through toggle so that tape enters and exits from the round back side of toggle.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Attach protective flap.						
	Position protective flap to left front opening, as indicated on pattern, and single stitch 3/16 to 1/4 inch from back edge of flap from the top of flap to and including crotch seam.	301	LSa-1	10-14	30-3	30-3	
10.	Attach labels.						
	a. Center (1 inch tolerance) the size label 1-1/2 ± 1/8 inches from top edge of waist at back seam and stitch on all four sides.	301	LSbj-1	10-14	30-3	30-3	
	b. Abut top edge of identification label to bottom edge of size label and stitch on all four sides. The top edge of the identification label may be caught in the bottom line of stitching of the size label.	301	LSbj-1	10-14	30-3	30-3	
	c. Stitch the instruction label to right center back (1 inch tolerance) on all four sides with the top edge of the label 1-1/2 ± 1/8 inches from top edge of waist.	301	LSbj-1	10-14	30-3	30-3	
	d. When a combination size, identification and instruction label is used, it shall be positioned as in "a" above.						
11.	Attach cuffs.						
	a. Cut cuff material 14 ± 1/4 inches long. Fold to double thickness.						
	b. Position top raw edges of cuff even with bottom edge of liner leg (leg inserted in cuff) and over-edge stitch together.	502 or 503	EFd-1	6-10	50-3 50-3	50-3 50-3	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Attach cuffs. (cont'd)						
	c. Single stitch cuff to liner 3/8 to 1/2 inch from raw edge.	301 or 401	LSq-2(a)	10-14	30-3 30-3	30-3 50-3	
	or						
	d. Stitch and overedge stitch cuff to liner leg in one operation with the bight of the overedge stitching 1/4 inch gage.	515 or	LSbm-3(a)	10-14	30-3 (chain stitching) 50-3 (overedge stitching)	50-3 50-3	
	e. Pull cuff to finished position and raise stitch liner through seam allowance 3/16 to 5/16 inch from seam.	301	LSq-2(b) or LSbm-3(b)	10-14	30-3	30-3	
	f. The cuffs shall finish 6 ± 1/4 inches in length.						
12.	Buttonholes.						
	a. Make a horizontal buttonhole in the center of waistband as indicated on pattern (1/4 inch off center tolerance permitted) with the inside edge of eyelet end 1/2 to 5/8 inch from left front edge.	Buttonhole		52-56 per buttonhole including tack	30-3	50-3	
	b. Make a horizontal buttonhole in the center of waistband as indicated on pattern (1/4 inch off center tolerance permitted) with the inside edge of eyelet end 3-3/4 ± 1/8 inches from right front edge.	Buttonhole		52-56 per buttonhole including tack	30-3	50-3	

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TABLE I. (cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Buttons.						
	a. Sew one button in the center of waistband 1 ± 1/4 inches in from right front edge to correspond to buttonhole on end of right waistband. (Measurement taken at center of button.)	101 or		14-16 per button	30-3		
		301		14-16 per button	30-3	30-3	
	b. Sew one button on protective flap to correspond to buttonhole in waistband.	101 or		14-16 per button	30-3		
		301		14-16 per button	30-3	30-3	
14.	Bartacking.						
	a. Place a horizontal bartack through front and protective flap with the front edge of bartack 2-1/2 ± 1/8 inches from right front opening and 7-3/4 ± 1/8 inches from top edge of waist.	Bartack		28 per bartack	30-3	30-3	
	b. Place a bartack diagonal to bottom of front opening catching the bottom edge of protective flap.	Bartack		28 per bartack	30-3	30-3	
15.	Clean liner.						
	a. Trim and remove all loose threads.						
	b. Remove all spots and stains.						
	c. Remove all size markings required by operation 3.						

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3.7 Finished measurements. The finished measurements shall be as shown in table II.

TABLE II. Finished measurements (in inches)

Size	1/2 Waist 1/	Inseam 2/
X-Small	14-1/2	42-3/4
Small	16-3/8	42-3/4
Medium	18-1/4	42-3/4
Large	20-1/8	42-7/8
X-Large	22-1/8	42-7/8
Tolerance	+ 3/4 - 1/2	± 3/4

1/ Measurement taken with the waist buttoned, the liner laid flat and smooth. Measure across front at the center of the waistband from side seam to side seam.

2/ Measure along inseam of liner from end of crotch to bottom edge of knitted cuff.

3.8 Workmanship. The finished liner shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document. In addition, testing shall be performed on components listed in table III for characteristics noted. Wherever applicable, tests shall be conducted in accordance with methods prescribed in FED-STD-191. All test reports shall contain the individual values used in expressing the final results. Sampling shall be in accordance with the following:

Lot size (lot unit)	Sample size (sample unit)
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The unit for expressing lot sizes and sample unit for testing each component shall be in accordance with applicable subsidiary documents and as follows:

Component	Lot size expressed in	Sample unit
Cotton sateen cloth	Yards	1/2 yard full width
Cotton knit tubing	Yards	1-1/2 continuous yards

TABLE III. Test methods

Component	Characteristic	Requirement paragraph	Test method
Cotton sateen cloth	Dynamic absorption	3.2.1	5500
Cotton knit tubing	Material identification	3.2.2	1200
	Type of knit	3.2.2	Visual 1/
	Yards per pound	3.2.2	5041
	Wales per inch	3.2.2	5070
	Courses per inch	3.2.2	5070
	Width (flat)	3.2.2	5020
	pH	3.2.2.1	2811
	Color	3.2.2.2	Visual 1/

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TABLE III. Test methods (cont'd)

Component	Characteristic	Requirement paragraph	Test method
Cotton knit tubing (cont'd)	Presence of labile sulfur	3.2.2.2	2020
	Colorfastness to:	3.2.2.3	
	- laundering		5610
	- perspiration		5680
	- crocking		5651
	Nonfibrous material starch and protein content (including chloroform-soluble and water-soluble material)	3.2.2.4	2611

1/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".

4.2.2 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of one liner. The sample unit shall be one liner. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for major and minor A combined defects, and 40.0 for total (major, minor A and minor B combined) defects.

Defects	Classification Major	Minor A	Minor B
Material defects and damages			
a. Applicable to cotton sateen cloth:			
1. Any hole, needle chew, cut, tear, mend, burn, or exposed drill hole.	X		
2. Any defect, such as a smash, multiple float, or slub.	X		

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Defects	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)			
b. Applicable to knitted cuffs:			
1. Hole, run, or drop stitch(es).	X		
2. Elasticity lacking, i.e., tight or stretched.		X	
3. Slubby yarn.			X
4. Unsightly mend.			X
NOTE: Material defects and damages are to be classified as indicated above only when the condition is one which weakens the cloth or liner or when it is so conspicuously located as to be clearly noticeable. Nonweakening conditions which are inconspicuous shall not be classified as a defect.			
Cleanness			
a. Any spot or stain on outside.			X
b. Thread ends not trimmed or loose thread ends not removed from liner (to be scored only when the condition exists on major portion of liner).			X
c. One or more size markings not removed, when applicable.			X
Components and assembly			
a. Any component part or required operation omitted (unless otherwise classified herein).	X		
b. Any operation not performed as specified (unless otherwise classified herein).		X	
Cutting			
Any component part not cut in accordance with directional lines indicated on pattern or not in accordance with specified requirements.	X		

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Defects	Classification Major	Minor A	Minor B
Seams and stitching			
a. Accuracy of seaming:			
1. Seams twisted, puckered, or pleated (unless otherwise classified herein).		X	
2. Part of liner caught in any unrelated operation or stitching.	X		
3. Ends of seams or stitching produced with 301 stitch type when not caught in other seams or stitchings backtacked less than 1/4 inch.			X
• 4. Stitching overlapping end(s) of thread break less than 1/2 inch.			X
5. Ends of a continuous line of stitching not overlapped or overlapped less than 1/2 inch.		X	
b. Gage of stitching (edge, raised, or double stitching):			
1. Unevenly gaged or various stitchings not uniformly gaged.			X
2. Beyond width specified or varies more than 1/16 inch when no range is specified.			X
c. Open seams - any open seam:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch but not more than 1 inch		X	
- more than 1 inch.	X		
NOTE: One or more broken or two or more continuous skipped or runoff stitches on joined seams constitutes an open seam. On double-stitched seams, a seam is considered open when either side is open.			
d. Raw edges:			
1. On outside:			
Any raw edge:			
- up to 3/16 inch inclusive		X	
- more than 3/16 inch.	X		

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Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
2. On inside: - any raw edge more than 1/4 inch (whenever applicable.		X	
NOTE: Raw edge not securely caught in stitching shall be classified as an open seam.			
e. Runoffs:			
1. Joining seams. When resulting in an open seam, use "open seam" classification.			
2. Edge or raised stitching (when not resulting in an open seam): - 1/4 to 1/2 inch inclusive			X
- more than 1/2 inch.		X	
f. Seam and stitch type:			
1. Wrong seam or stitch type.		X	
2. Looper thread of 401 stitch type finishing on outside of liner, except on turn back of joining seams.	X		
g. Stitch tension:			
1. Loose tension, resulting in a loose seam.	X		
2. Loose tension of edge or raised stitching, resulting in loosely exposed bobbin thread or top thread.		X	
3. Tight tension (stitches break when strain is applied in the direction of seam.)	X		
h. Stitches per inch (to be scored only when the condition exists on major portion of seam):			
1. Less than specified: - one to three stitches inclusive			X
- four stitches		X	
- more than four stitches.	X		

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Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
2. One or more stitches more than specified.			X
i. Stitches skipped or broken (when seam is seamed, turned, and stitched):			
On edge stitching of waistband:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch.		X	
Buttonholes (construction)			
a. One or more omitted or added.	X		
b. Stitching not securely caught in cloth, causing stitching to pull away from fabric.	X		
c. Gimp omitted:			
- one buttonhole			X
- two or more buttonholes.		X	
d. Ragged edges.			X
e. Wrong type or uncut buttonhole.		X	
f. Ends not securely tacked:			
- one buttonhole			X
- two or more buttonholes.		X	
g. One or more broken or skipped stitches in one or more buttonholes or eyelets.		X	
h. Purling on the inside of liner.			X
i. Cut length less than 3/4 or more than 7/8 inch.		X	
j. Eyelets less than 1/8 or more than 3/16 inch in diameter.			X
Buttons			
Missing, loose, or defective:			
- one button		X	
- two buttons.	X		

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	Classification		
	Major	Minor	
		A	B
Bartacks			
Missing, insecure, or misplaced, not serving intended purpose.			
- one bartack			X
- two or more bartacks.		X	
Crotch seam			
a. Left front not overlapping, or overlapping seam of right front by more than 1/4 inch at base of opening.		X	
b. Seam not centered with seat seam by more than 3/8 inch (center to center).		X	
Seat seam			
Right back not overlapping left back.		X	
Inseams and outseams			
Front parts not overlapping back parts.		X	
Front opening			
Opening uneven in length by:			
- 1/4 inch up to 3/8 inch inclusive			X
- more than 3/8 inch.		X	
Waist			
a. Construction:			
1. Twisted.		X	
2. Width less than 1-1/8 or more than 1-3/8 inches.			X
3. Irregular in width by more than 1/4 inch.		X	
4. Hem finishing toward the inside.		X	

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	Classification		
	Major	Minor A	Minor B
Protective flap			
a. Button out of alignment with buttonhole at waist.			X
b. Bartack at base of front opening not through protective flap.		X	
c. Front edge of bartack less than 2-3/8 or more than 2-5/8 inches from edge of opening; or less than 7-5/8 or more than 7-7/8 inches from top edge of waist.		X	
Trouser bottom(s)			
a. Cuff finishing less than 5-3/4 or more than 6-1/4 inches in length.			X
b. Cuff badly twisted.			X
c. Cuff spliced (not tubular).	X		
d. Not constructed of double thickness.	X		
Labels			
a. Size label, instruction label, and combination label (when applicable):			
1. Missing, incorrect, or illegible.	X		
2. Not positioned as specified, or not stitched on all four sides.			X
3. Positioned off center by more than 1 inch.			X
b. Identification label (when applicable):			
1. Missing, incorrect, or illegible.			X
2. Not positioned as specified, or not stitched on all four sides.			X

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	Classification		
	Major	Minor A	Minor B
Waistband buttonholes			
a. Inside edge of eyelet end of waist buttonhole less than 1/2 or more 5/8 inch from front edge.			X
b. Inside edge of eyelet end of protective flap buttonhole less than 3-5/8 or more than 3-7/8 inches from top edge of waistband.			X
c. Waistband buttonholes off center of waistband by more than 1/4 inch.			X
d. Button positioned on waistband less than 3/4 or more than 1-1/4 inches from front edge (measurement taken at center of button.)			X
Adjustment tape buttonholes or eyelets			
a. Reinforcement piece omitted.		X	
b. One or both buttonholes or eyelets not fully constructed through reinforcement piece.			X
c. Eyelet end of buttonhole positioned toward back of liner.			X
d. Buttonhole not positioned horizontally.			X
Waist adjustment tape			
a. Extending more than 4 or less than 2 inches beyond buttonhole or eyelet.			X
b. Bartack omitted at center back of waistband.		X	
c. Adjustment tape caught in waistband stitching.	X		

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4.2.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in 3.7. Any dimension deviating from the specified requirement and inseams uneven in length by 1/2 inch or more shall be classified as defects. The lot size shall be expressed in units of one liner. The sample unit shall be one liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.2.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged except that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container. Improperly folded liners, inadequately tied bundles, contents improperly covered with paper.
Content	Number of bundles per container is more or less than required. Number of liners per bundle is more or less than required. 1/ Size shown on one or more liners not as specified on container. 1/

1/ For this defect, one bundle from each container in the sample shall be examined.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Each buttoned liner shall be laid flat and folded in half along the length with side seams meeting. The liner shall be folded from the leg bottom up toward the waistband to the overall length of approximately 23 inches. For the larger sizes, it may be necessary to turn under the cuffs to achieve the desired folded length. Five pairs of folded liners of one size only shall be stacked in a bundle, with the waistbands alternated, and securely tied at each end with cotton tape or twine. The bundle shall then be securely tied with cotton tape or twine.

5.1.2 Commercial. The liners shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Forty liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, eight in depth within a shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2 Level B packing. Forty liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, eight in depth within a shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

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* 5.2.3 Commercial packing. Liners, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side, directly under the printing or stenciling a white paper label 5- by 4-inches with the words "MIXED SIZES" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The liner is intended for wear by military personnel of the Department of Defense as an inner garment to provide protection against chemical warfare agents.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Size(s) required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Sample. For access to sample of the end item, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document.

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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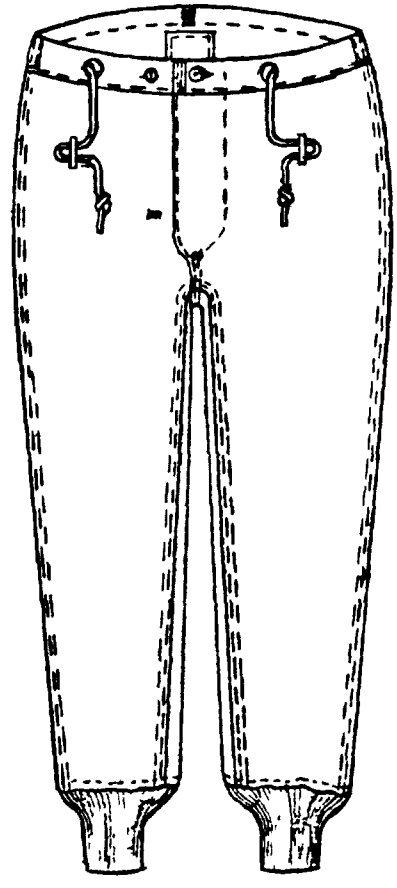
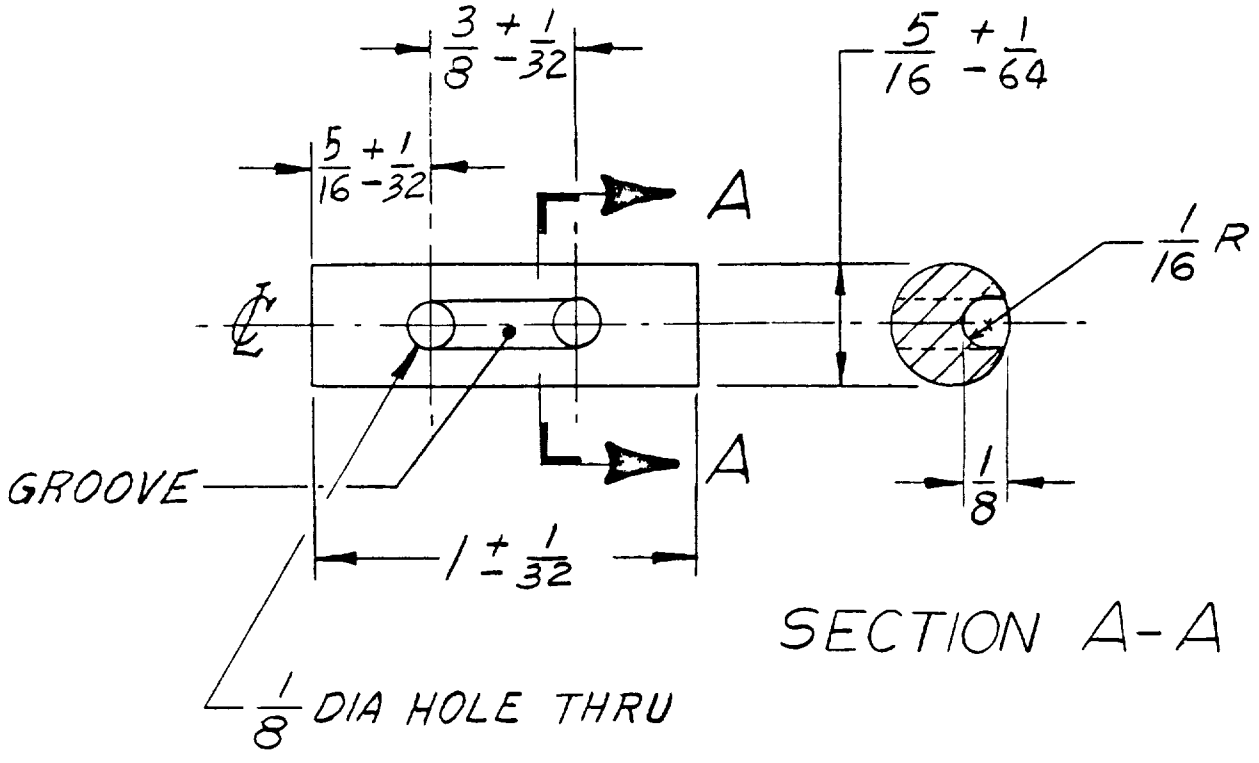


FIG. 1 LINER, TROUSERS, CHEMICAL PROTECTIVE

2-1-825 C

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NOTE:-

1. REMOVE ALL BURRS AND SHARP EDGES.

FIGURE 2. TOGGLE, LINER, TROUSERS, CHEMICAL PROTECTIVE

2-1-1061

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